Chapter 13
Binomial Trees

Chapters 10 to 12 described what options are, what relationships they must satisfy, and how they are combined. None of them priced one. Chapter 11 established bounds, and bounds are not a price. This chapter supplies the missing element, and it does so with almost no mathematics.

The tree here is a teaching device before it is a computational method. It is used because it is the simplest possible market in which the entire argument of option pricing can be seen at once (one period, two states, two traded instruments), and because in that setting every step can be checked by hand. How to build trees efficiently, how many steps they need, and how to make them converge quickly are questions about computation, and they belong to Chapter 20.

What the device makes visible is the argument by which a derivative acquires a price at all: **replication**, **no arbitrage**, and **risk-neutral valuation**. That argument is the foundation of everything in Part III, and it is genuinely difficult to motivate once the machinery of continuous time is in the way. Here it fits on a page.

Sections 13.1 to 13.4 are the argument in its smallest setting: one step, risk-neutral valuation, state prices, and why two states suffice. Sections 13.5 to 13.7 extend it to many steps, to the general form, and to American exercise. Sections 13.8 to 13.10 are the choices that connect the tree to a real underlying (the up and down moves, convergence, and dividends), and Sections 13.11 and 13.12 close with the identity of pricing and hedging and with what the tree assumes.

Later chapters depend on this one. Chapters 15 and 16 take risk-neutral valuation as given; Chapter 18 builds on the identity of pricing and hedging established in Section 13.11; Chapter 19 works in the incomplete markets that Section 13.4 introduces.

13.1 A ONE-STEP BINOMIAL MODEL

Suppose a stock trades at \$20 and that in three months it will be worth either \$22 or \$18. Nothing else can happen. We want the value f of a European call struck at \$21, which pays \$1 in the up state and nothing in the down state. Figure 13.1 carries both on one tree (the stock's two possible prices above, the call's two possible values below), and the rest of this chapter shows how the stock's two prices force the call's value today.

The method is to construct a portfolio whose value is the same in both states. Hold Δ shares and write one call. In the up state the portfolio is worth $22\Delta - 1$; in the down state, 18Δ. These are equal when

$$22\Delta - 1 = 18\Delta \quad \Longrightarrow \quad \Delta = 0.25$$

At $\Delta = 0.25$ the portfolio is worth $18 \times 0.25 = \$4.50$ in *either* state. It is riskless. A riskless portfolio must earn the risk-free rate, or there is an arbitrage, so its value today is $4.50\, e^{-rT}$. With $r = 4\%$ and $T = 0.25$,

$$4.50\, e^{-0.04 \times 0.25} = 4.50\, e^{-0.01} = \$4.455$$

The portfolio today costs $20\Delta - f = 5 - f$, so

$$5 - f = 4.455 \quad \Longrightarrow \quad f = \$0.545$$

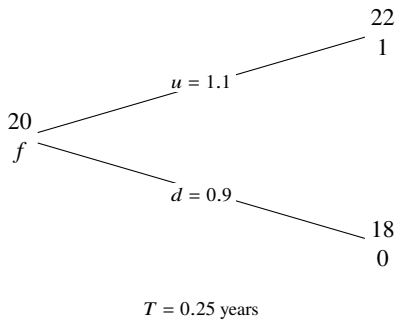


Figure 13.1: Stock price above, call value below. $K = 21$.

Read the same construction the other way and it is a recipe. Holding 0.25 shares and owing \$4.455 is worth $0.25 \times 22 - 4.50 = \$1$ in the up state and $0.25 \times 18 - 4.50 = \$0$ in the down state: it *replicates* the call. Its cost is $5 - 4.455 = \$0.545$, and that is the price: the call is worth what it costs to manufacture. Example 13.1 shows what happens if it trades at any other price.

Example 13.1 Checking the arbitrage

Suppose the call traded at \$0.60 rather than \$0.545. Write the call, buy 0.25 shares:

$$\text{cash out} = 0.25 \times 20 - 0.60 = \$4.40$$

Borrow the \$4.40 at 4%. In three months repay $4.40\, e^{0.01} = \$4.444$, and the portfolio is worth \$4.50 whichever way the stock moves. The profit is \$0.056 in every state, from nothing invested—and \$0.056 is $(0.60 - 0.545)e^{0.01}$, the mispricing carried forward, as in Example 11.1. If instead the call traded at \$0.50, run the trade in reverse: buy the call, short 0.25 shares, invest the proceeds.

Notice what never entered the calculation: no one was asked how likely the up state is, or what return investors require for bearing risk. The argument works for a trader who thinks the stock will certainly rise and for one who thinks it will certainly fall. That is the feature to hold on to, because Section 13.2 shows it is not an accident of this example.

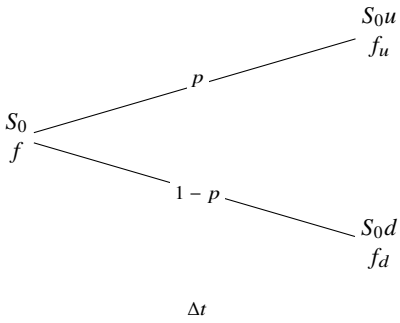


Figure 13.2: The general one-step tree. Stock price above, derivative value below; p and $1 - p$ are the risk-neutral probabilities of (13.2).

The general one-step formula. Write the stock as moving from S_0 to S_0u or S_0d over a step of length Δt, with the derivative paying f_u or f_d, as in Figure 13.2. The same construction gives the number of shares held per option, the hedge ratio known as the option's delta,

$$\Delta = \frac{f_u - f_d}{S_0u - S_0d} \tag{13.1}$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

and, after substituting and rearranging,

$$\boxed{f = e^{-r\,\Delta t}\,[p f_u + (1-p) f_d]} \qquad \text{where} \qquad p = \frac{e^{r\,\Delta t} - d}{u - d} \tag{13.2}$$

For no arbitrage we need $d < e^{r\Delta t} < u$; otherwise the stock always beats or always loses to cash. That condition also guarantees $0 < p < 1$.

Check the example:

$$p = \frac{e^{0.04\times 0.25} - 0.9}{1.1 - 0.9} = 0.5503,$$
$$f = e^{-0.04\times 0.25}\,[0.5503 \times 1 + 0.4497 \times 0] = \$0.545,$$

as before. Here and throughout, intermediate values are carried unrounded and printed to four places, so the last digit of a product can differ from the product of the printed factors.

13.2 RISK-NEUTRAL VALUATION

Equation (13.2) has the form of a discounted expected payoff, with p playing the role of a probability. The meaning of p needs care, because most of the confusion in derivative pricing starts here.

What p is. Compute the expected stock price using p:

$$\mathbb{E}[S_T] = pS_0u + (1-p)S_0d = S_0\,[pu + (1-p)d] = S_0e^{r\,\Delta t}$$

(the last step is algebra: substitute p and simplify). So p is the probability under which the stock is expected to earn the risk-free rate. It is called the **risk-neutral probability**, and the corresponding expectation is taken in the **risk-neutral world**.

What p is not. It is not the real-world probability of the stock rising, and it is not anyone's forecast. In the example $p = 0.5503$; the true probability might be 0.7 or 0.3, and the option price would be \$0.545 either way. This is exact, not an approximation: the true probability is irrelevant.

Why this works. The reason is not that anyone is risk-neutral. It is that the option can be *replicated*: Section 13.1 built a portfolio of stock and cash that reproduces the option's payoff exactly. Two things with identical payoffs must have identical prices, whatever anyone's attitude to risk. Attitudes to risk do set the interest rate and the price of the underlying asset, but once those two prices are given they have no further say in the option's. Preferences and real-world probabilities drop out of the problem, and the only assignment of probabilities that survives (the only *measure*, in the language of Part III) is the one under which no risk premium is required, which is p.

The result generalizes into the principle underlying all of Part III:

> **Risk-neutral valuation.** The value of a derivative is the expected value of its payoff in a world where all assets earn the risk-free rate, discounted at the risk-free rate.

The risk-neutral probability is a tool for calculating derivative prices, not a description of the world. We do not believe investors are risk-neutral. We use the risk-neutral measure because it gives the right answer to a question about relative prices, and it gives it far more easily than the alternative. Chapter 15 shows the same principle at work in continuous time; Chapter 28 shows what happens when one changes to a different measure entirely. Example 13.2 shows what the alternative involves.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 13.2 The real-world calculation, for comparison

Suppose the true probability of the up state is 0.7. Then the expected stock price is $0.7\times22+0.3\times18 = \$20.80$, an expected return of

$$\frac{20.80}{20} - 1 = 4\% \text{ over three months}$$

The expected call payoff is $0.7 \times 1 + 0.3 \times 0 = \$0.70$. To discount it we need the return investors require *on the option*, which is not 4%: the option is a leveraged position in the stock and is far riskier.

That discount rate cannot be looked up. It depends on how leveraged the option is, and the leverage depends on the option's price, which is what we were trying to find. The real-world approach is circular unless a preference model is supplied from outside.

Run backward from the known answer, and the circle closes. Discounting \$0.70 back to the known price \$0.545 implies a required return of $0.70/0.5448 - 1 = 28.5\%$ per quarter, an enormous risk premium, correctly reflecting how leveraged the option is. Risk-neutral valuation sidesteps this entirely. The required return on the option, which no one can observe, never has to be found: the replication argument makes the risk premium cancel out, and what remains is the risk-free rate, which everyone can observe.

13.3 STATE PRICES AND THE ABSENCE OF ARBITRAGE

Equation (13.2) can be read a second way, and the second reading connects this chapter to Chapters 11 and 12 and states the principle in its most general form. Group the discount factor with the probabilities:

$$f = \underbrace{e^{-r\Delta t}p}_{\psi_u} f_u + \underbrace{e^{-r\Delta t}(1-p)}_{\psi_d} f_d \tag{13.3}$$

The two quantities ψ_u and ψ_d are **state prices**. Each is the price, today, of a security that pays \$1 if a particular state occurs and nothing otherwise. Such a security is called an **Arrow–Debreu security**, and once the price of each is known, *every* derivative on this underlying can be priced by multiplying its payoffs by the corresponding state prices and adding. In the running example,

$$\psi_u = e^{-0.04\times0.25} \times 0.5503 = 0.5448,$$
$$\psi_d = e^{-0.04\times0.25} \times 0.4497 = 0.4453,$$

and Example 13.3 confirms both by replication.

Example 13.3 Building and using state prices

Verify that $\psi_u = 0.5448$ really is the price of \$1 in the up state. Replicate that payoff with Δ shares and B in cash:

$$22\Delta + Be^{0.04\times0.25} = 1, \qquad 18\Delta + Be^{0.04\times0.25} = 0$$

Subtracting gives $\Delta = (1-0)/(22-18) = 0.25$, which is (13.1) with the payoffs 1 and 0, and then $Be^{0.04\times0.25} = -4.50$, so $B = -\$4.4552$. The cost today is

$$0.25 \times 20 - 4.4552 = \$0.5448$$

which is ψ_u, as claimed. In this example the security paying \$1 up and \$0 down *is* the call struck at 21, which is why $f = \psi_u$. Similarly, replicating \$1 in the down state takes $\Delta = -0.25$ and $B = \$5.4453$, so $\psi_d = -0.25 \times 20 + 5.4453 = \$0.4453$.

Now price something else with no further work. A claim paying \$3 in the up state and \$7 in the down state is worth

$$3 \times 0.5448 + 7 \times 0.4453 = 1.6343 + 3.1169 = \$4.7512$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Checking by replication, (13.1) gives $\Delta = (3-7)/(22-18) = -1$, then $B = 25e^{-0.04\times0.25} = \$24.7512$, and the cost is $-20 + 24.7512 = \$4.7512$. Two state prices value every derivative on this stock: the replication is done once, and every payoff afterward is a weighted sum.

Note also that $\psi_u + \psi_d = e^{-r\Delta t} = 0.99005$. A claim paying \$1 in *both* states is a riskless bond, so the state prices must sum to the discount factor, which is where the discounting in (13.2) comes from, and why p and $1-p$ behave like probabilities: they are state prices *normalized* by the bond price.

The arbitrage-free condition. State prices make precise what "no arbitrage" requires. A security paying \$1 in one state and nothing otherwise cannot have a negative price (the buyer would be paid to take a payoff that is never negative), and it cannot be free, since it sometimes pays. So both state prices must be strictly positive, which given $\psi_u + \psi_d = e^{-r\Delta t}$ is exactly the condition

$$0 < p < 1 \quad \Longleftrightarrow \quad d < e^{r\Delta t} < u$$

met earlier in Section 13.1.

This is a special case of a general result:

> **The market admits no arbitrage if and only if a set of strictly positive state prices exists**; equivalently, if and only if there is a risk-neutral measure, a probability measure that gives every state a positive probability and under which every asset's expected return is the risk-free rate.

Read in that direction, risk-neutral valuation stops looking like a trick. The risk-neutral measure is not an assumption added to the model; it is another name for the absence of arbitrage. A market with no arbitrage always has such a measure, and a market with an arbitrage never does.

The connection to Chapter 12. Section 12.8 built a digital option as the limit of a tight call spread. Push that construction one derivative further: a butterfly of vanishing width centered at K pays only if the underlying finishes at K, and its price, divided by the square of its width, tends to $\partial^2 c/\partial K^2$. A butterfly of that kind *is* an Arrow–Debreu security, and its price *is* a state price. The risk-neutral density of the terminal price, which Chapter 19 reads off a continuum of traded option prices, and the state prices of this section are the same object seen from two directions, one observed in the market, the other constructed from a replication argument. Chapter 19 uses the first; most of the rest of Part III uses the second.

13.4 WHY TWO STATES? COMPLETENESS

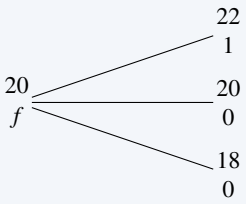
An obvious objection has been left standing since Section 13.1: a stock does not have two possible prices in three months. Section 13.9 answers part of that by letting the steps shrink, but the deeper answer is not about realism at all, and it is the most important idea in this chapter.

The two-state assumption is what makes the market **complete**: every payoff can be replicated by trading the instruments available. There are two traded instruments (the stock and the bond) and two states, so the replicating equations are two equations in two unknowns and have a unique solution. *Every* payoff can therefore be manufactured, so every payoff has a price. Add a third state and that stops being true, as Example 13.4 shows.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 13.4 Three states, and the price is no longer unique

Suppose the stock at \$20 can move to \$22, \$20, or \$18. Everything else is as before: $r = 4\%$, $\Delta t = 0.25$, and a call struck at \$21 paying \$1, \$0, \$0.



Try to replicate it with Δ shares and B in cash:

$$22\Delta + Be^{0.01} = 1, \qquad 20\Delta + Be^{0.01} = 0, \qquad 18\Delta + Be^{0.01} = 0$$

The last two give $2\Delta = 0$, so $\Delta = 0$ and then $B = 0$, which contradicts the first. The call cannot be replicated, and the argument that produced a price collapses.

What survives? The state prices must still be positive and must still price the two traded instruments correctly:

$$22\psi_1 + 20\psi_2 + 18\psi_3 = 20, \qquad \psi_1 + \psi_2 + \psi_3 = e^{-0.01} = 0.99005$$

Two equations in three unknowns have a whole family of solutions rather than one. Eliminating ψ_2 gives $\psi_1 = \psi_3 + 0.0995$ and then $\psi_2 = 0.890548 - 2\psi_3$, so requiring $\psi_2 > 0$ and $\psi_3 > 0$ confines ψ_3 to the open interval $(0,\ 0.445274)$.

The call's value is ψ_1, so it can be anything in

$$\$0.0995 \ < \ f \ < \ \$0.5448$$

Every price in that interval is consistent with no arbitrage. The theory no longer delivers a price—it delivers bounds, and the upper one is more than five times the lower.

Note what sits at the two ends. As ψ_3 approaches 0.445274, ψ_2 goes to zero, the middle state is priced at nothing, and the value approaches \$0.5448, the binomial answer of Section 13.1. As ψ_3 approaches zero instead, the down state is priced at nothing and the value approaches \$0.0995, the binomial answer for a tree whose two moves are to \$22 and \$20. Each bound is a two-state model, the limiting case of the three-state one.

This example is worth dwelling on because it identifies precisely what the binomial assumption is doing. The assumption is no claim about how stocks behave; it makes the number of states equal the number of traded instruments, and that equality, not realism, is what converts Chapter 11's bounds into a price.

The general statement is the second half of the result quoted above:

> **The market admits no arbitrage and is complete if and only if there is exactly one risk-neutral measure**; equivalently, if and only if there is exactly one set of strictly positive state prices. Where it is unique, every derivative has one arbitrage-free price. Where it is not, derivatives have a range of arbitrage-free prices, and selecting one requires something beyond arbitrage: a model, or a view about preferences.

This is not a technicality confined to contrived examples. Real markets are incomplete for exactly this reason: prices jump, volatility moves unpredictably, and the number of things that can happen exceeds the number of instruments available to hedge them. The consequence appears throughout the rest of the book. It is why Chapter 19's volatility smile (options at different strikes implying different volatilities for the same underlying) cannot be argued away, why Chapter 34's exotic payoffs carry model risk that vanilla options do not, and why two desks using different models can quote different prices for the same structure without either committing an arbitrage.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Completeness is the assumption that lets this chapter say "the price." Everywhere it fails, the accurate phrase is "a price, given a model."

13.5 TWO-STEP TREES AND BACKWARD INDUCTION

With more than one period, apply (13.2) repeatedly, working backward from expiration one step at a time, the procedure called **backward induction**. Figure 13.3 extends the example to two three-month steps.

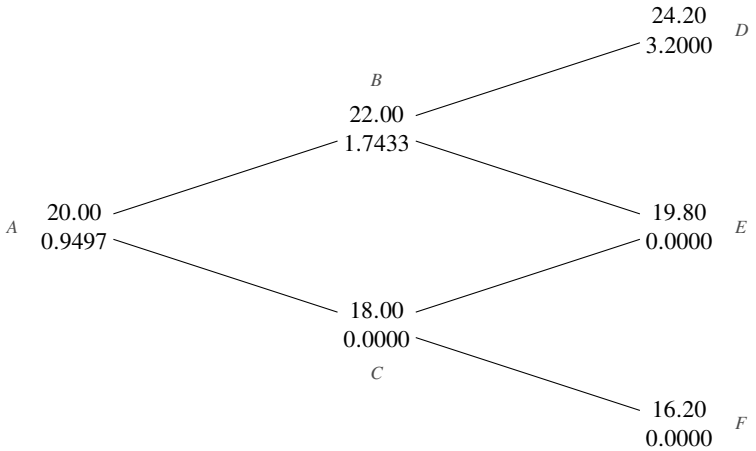


Figure 13.3: Stock price above, European call value below. $K = 21$, $r = 4\%$.

At the terminal nodes D, E, and F the call is worth $\max(S_T - 21, 0)$: \$3.20, \$0, \$0. Because u, d, and r are the same at every step, $p = 0.5503$ is the same at every node too. Working back with it and $e^{-r\Delta t} = 0.99005$:

$$\text{node B}: \quad e^{-0.01}\,[0.5503 \times 3.20 + 0.4497 \times 0] = \$1.7433$$

$$\text{node C}: \quad e^{-0.01}\,[0.5503 \times 0 + 0.4497 \times 0] = \$0$$

$$\text{node A}: \quad e^{-0.01}\,[0.5503 \times 1.7433 + 0.4497 \times 0] = \$0.9497$$

Two features of the procedure matter more than the arithmetic.

The tree recombines. An up-then-down move and a down-then-up move both give $S_0ud = 19.80$, so the number of nodes grows linearly rather than exponentially: $n + 1$ nodes after n steps rather than 2^n. This is what makes trees computationally practical, and Section 13.10 shows what happens when it fails.

The replication is dynamic. The riskless portfolio of Section 13.1 must be *rebalanced* at each node, because Δ from (13.1) differs at each one. At node B, $\Delta = (3.20 - 0)/(24.20 - 19.80) = 0.7273$; at node C, $\Delta = 0$. Pricing the option assumes the hedger can and does trade at every step. Section 13.12 returns to what that assumption costs.

Example 13.5 Following the hedge through the tree

Take the tree of Figure 13.3 and follow a trader who writes the call and hedges it.

From (13.1) the deltas are $\Delta_A = (1.7433 - 0)/(22 - 18) = 0.4358$ at node A, $\Delta_B = (3.20 - 0)/(24.20 - 19.80) = 0.7273$ at node B, and $\Delta_C = 0$ at node C. The arithmetic below carries the deltas unrounded.

Today (node A). Sell the call for \$0.9497. Buy 0.4358 shares at \$20, costing \$8.7164. Borrow the difference:

$$8.7164 - 0.9497 = \$7.7667$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Three months on. The debt has grown to $7.7667\,e^{0.01} = \$7.8448$. If the stock is at 22 (node B), the shares are worth $0.4358 \times 22 = \$9.5881$, so the position is worth

$$9.5881 - 7.8448 = \$1.7433$$

which is exactly the call's value at node B. If instead the stock is at 18 (node C), the shares are worth \$7.8448 and the position is worth precisely nothing, again the call's value there.

Rebalancing at node B. Delta must rise from 0.4358 to 0.7273, so buy 0.2915 more shares at \$22, costing \$6.4119, financed by borrowing. Total debt becomes 7.8448 + 6.4119 = \$14.2567.

At expiration. The debt grows to $14.2567\,e^{0.01} = \$14.4000$. If the stock reaches 24.20, the shares are worth $0.7273 \times 24.20 = \$17.6000$ and the position is worth

$$17.6000 - 14.4000 = \$3.2000$$

which is exactly the call's payoff. If the stock falls to 19.80, the shares are worth \$14.4000, the position is worth zero, and the call expires worthless.

The hedge closes to the penny in every state, having been financed entirely by the \$0.9497 received at the outset. That is what it means for \$0.9497 to be the price—it is the cost of manufacturing the option's payoff.

Two things in the walkthrough deserve emphasis. The trader bought more shares at 22 and would have sold shares at 18: the hedge *buys high and sells low* by construction, and the accumulated cost of doing so is precisely the option premium. And nothing was ever at risk (no step required a view), which is the replication argument made operational rather than asserted.

13.6 THE MODEL IN GENERAL FORM

The examples have been worked one node at a time. Set out formally, the model is compact, and its structure becomes visible in a way node-by-node arithmetic hides.

The lattice. Divide the derivative's life, T years, into n steps of length $\Delta t = T/n$. The underlying starts at S_0 and at each step is multiplied by u or by d. After i steps it has taken j up moves and $i - j$ down moves, so it stands at

$$S_{i,j} = S_0\, u^j d^{\,i-j}, \qquad j = 0, 1, \ldots, i \tag{13.4}$$

Because u and d multiply, only the count j matters, not the order of the moves. That is the recombining property, and it is why step i has $i + 1$ nodes rather than 2^i.

The measure. Under the risk-neutral measure Q, each step is an up move with probability p from (13.2), independently of the others. The number of up moves in n steps is therefore *binomially distributed*, which is the model's name:

$$Q(j \text{ up moves}) = \binom{n}{j} p^j (1-p)^{n-j}$$

The pricing formula. For a European claim paying $h(S_T)$ at maturity, backward induction through the whole tree collapses to a single sum:

$$f_0 = e^{-rT} \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j}\, h\big(S_0 u^j d^{\,n-j}\big) \tag{13.5}$$

and for a European call, $h(S) = \max(S - K, 0)$, so the terms with $S_0 u^j d^{n-j} \le K$ vanish and the sum runs only over the nodes finishing in the money.

Equation (13.5) is the two-step calculation of Section 13.5 written out in one line. Applying it there with $n = 2$ reproduces \$0.9497 directly, without the intermediate node values, though those values are exactly what an American option needs, which is why Section 13.7 cannot use this shortcut.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

The martingale property. There is a more revealing way to state what Q does. Under it, the *discounted* price of any traded asset is a **martingale**, meaning that its expected future value, discounted, equals its value today:

$$e^{-r\Delta t}\,\mathbb{E}^{Q}[S_{i+1} \mid S_i] = S_i, \qquad e^{-r\Delta t}\,\mathbb{E}^{Q}[f_{i+1} \mid S_i] = f_i \tag{13.6}$$

The first equality is the calculation of Section 13.2 restated; the second is equation (13.2) restated. That both hold under the same measure is the entire content of risk-neutral valuation, and it is the form in which the principle carries over to continuous time in Chapter 15 and to other numeraires (other choices of the asset in which prices are measured) in Chapter 28.

The continuous limit. With the parameter choice (13.8), Section 13.9 shows that as $n \to \infty$ the terminal distribution becomes lognormal (the logarithm of the price is normally distributed) and the sum (13.5) for a European call converges to the Black–Scholes–Merton formula, which Chapter 15 derives directly and which is the continuous-time counterpart of everything in this chapter.

13.7 AMERICAN OPTIONS

Trees remain in use because they handle early exercise, which Chapter 11 showed matters for puts and for calls on dividend-paying stocks, and which, outside a few special cases, no closed-form formula can address. The modification is small. At each node, compute the value of continuing (equation (13.2) applied to the two successor nodes) and compare it with the value of exercising immediately. The node's value is the greater:

$$f = \max\Big(\underbrace{e^{-r\Delta t}\,[p f_u + (1-p) f_d]}_{\text{continue}},\ \underbrace{\text{intrinsic value}}_{\text{exercise}}\Big) \tag{13.7}$$

Example 13.6 applies the comparison to a two-step put. The same comparison handles any decision embedded in a security, including a convertible bond's two rights, the holder's to convert and the issuer's to call (redeem early): each is a max or a min at the nodes where it can be exercised, and because the two interact node by node, the bond and its embedded options cannot be valued separately and added.

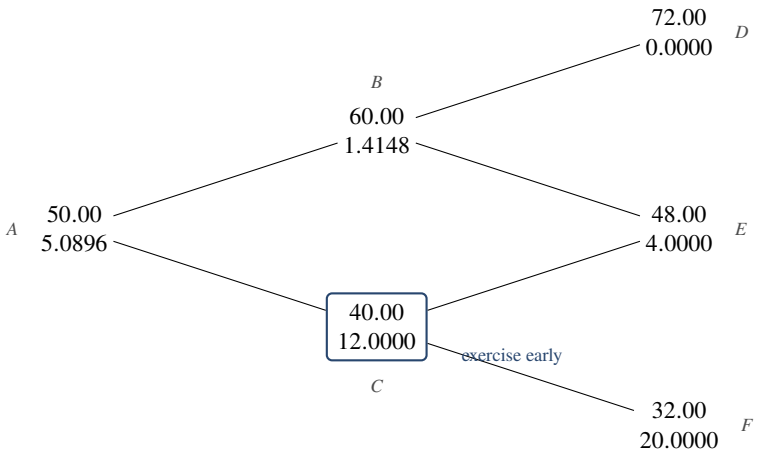


Figure 13.4: Stock price above, American put value below. $K = 52$, $r = 5\%$.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 13.6 An American put, and what early exercise is worth

A stock trades at \$50, with $u = 1.2$ and $d = 0.8$ over each of two one-year steps. Consider an American put struck at \$52, with $r = 5\%$.

First, $e^{0.05\times 1} = 1.051271$ and

$$p = \frac{1.051271 - 0.8}{1.2 - 0.8} = \frac{0.251271}{0.4} = 0.6282$$

Figure 13.4 sets out the tree. At the terminal nodes D, E, and F the put is worth \$0, \$4, and \$20.

Node B ($S = 60$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 0 + 0.3718 \times 4] = \$1.4148$$

Exercising yields $\max(52 - 60, 0) = 0$. Hold. Value \$1.4148.

Node C ($S = 40$). Continuation value is

$$e^{-0.05}\,[0.6282 \times 4 + 0.3718 \times 20] = \$9.4639$$

Exercising yields $52 - 40 = \$12$. **Exercise.** Value \$12.

Node A (today). Continuation value is

$$e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 12] = \$5.0896$$

Exercising yields $52 - 50 = \$2$. Hold. Value \$5.0896.

Now price the European put for comparison. The only difference is at node C, where the European put is worth its continuation value of \$9.4639 rather than \$12:

$$p_{\text{Eur}} = e^{-0.05}\,[0.6282 \times 1.4148 + 0.3718 \times 9.4639] = \$4.1927$$

The early-exercise premium is $5.0896 - 4.1927 = \$0.897$, more than 21% of the European value. Section 11.9 argued that an American put is worth more than its European counterpart; here is the gap, computed.

The tree also *locates* the exercise decision, which is its practical advantage. Exercise is optimal at node C and not at nodes A or B, so the critical price S^* of Figure 11.5 lies between 40 and 60 at that date, and below 50 today. Figure 13.5 traces it date by date on a 1,000-step tree built as Section 13.8 describes, with $\sigma = \ln(1.2/0.8)/2 = 20.3\%$, half the gap between the two-step tree's log moves in a year; the curve $S^*(t)$ it draws is the put's **early-exercise boundary**. The boundary starts just above 40, rises to about 42 by the one-year date the two-step tree bracketed, and climbs the rest of the way to the strike only in the final weeks.

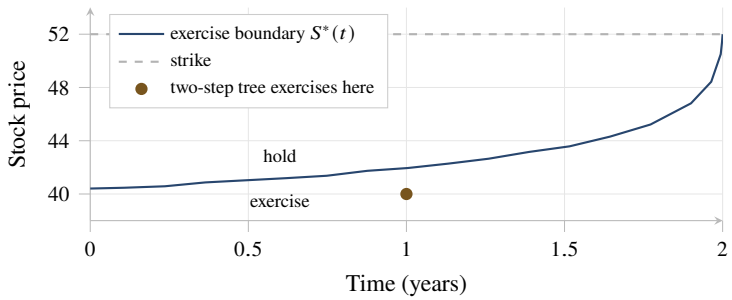


Figure 13.5: Early-exercise boundary of the Example 13.6 put on a 1,000-step tree, $\sigma = 20.3\%$.

13.8 CHOOSING u AND d

So far u and d have been given. In practice they are chosen so that the tree reproduces the volatility of the underlying, which is the only parameter of the model that must be estimated.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

The standard choice, due to Cox, Ross, and Rubinstein, is

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}} \tag{13.8}$$

where σ is the stock's **volatility**, the annualized standard deviation of its continuously compounded return, and Δt is the length of a step in years.

These match the variance of the stock's return over each step to $\sigma^2\Delta t$, to first order in Δt, which is the same variance the continuous-time model of Chapter 15 assumes. Setting $d = 1/u$ also centers the tree, so that an up move followed by a down move returns exactly to the starting price.

This leaves σ itself, the one input that cannot be read off a screen. It is the same number in the real world and in the risk-neutral one: moving from the true probability to p changes the tree's drift but, to first order in Δt, not its variance, so a volatility measured from historical prices can go into the tree unchanged. Estimating it, from history or by inverting a traded option's price, is taken up once, in Chapter 15; here σ is given, and Example 13.7 builds a tree from it.

Example 13.7 Building a tree from a volatility

A stock has volatility 30% a year. Build a tree with monthly steps, $\Delta t = 1/12$:

$$u = e^{0.30\sqrt{1/12}} = e^{0.086603} = 1.09046, \qquad d = 1/1.09046 = 0.91704$$

With $r = 4\%$,

$$p = \frac{e^{0.04/12} - 0.91704}{1.09046 - 0.91704} = 0.49762$$

One sanity check is worth performing every time: $d < e^{r\Delta t} < u$, equivalently $0 < p < 1$. Here $0.917 < 1.0033 < 1.090$. It fails when the step is too long for the volatility and rate, and the symptom is a negative probability.

Note also how little p moved from one-half. Expanding $e^{r\Delta t}$, u, and d to order Δt gives

$$p \approx \frac{1}{2} + \frac{1}{2}\left(\frac{r - \sigma^2/2}{\sigma}\right)\sqrt{\Delta t} = \frac{1}{2} + \frac{1}{2} \times \frac{0.04 - 0.3^2/2}{0.30} \times 0.2887 = 0.4976,$$

the exact value to four places. So the risk-neutral probability is close to a coin flip and the drift enters only as a small correction. The volatility sets u and d, and with them how far the tree spreads at each step.

13.9 CONVERGENCE

A two-step tree is a crude description of a stock. The model is justified because refinement removes the crudeness: as the number of steps rises, the tree value settles down to a limit, the Black–Scholes–Merton value that Chapter 15 derives. Here the point is only that the limit exists, so that the arbitrary-looking choice of two branches does not determine the answer.

Figure 13.6 shows the characteristic behavior, and two features of it matter in practice.

Convergence is slow. The error decays roughly as $1/n$, so halving it requires doubling the number of steps, and since a tree of n steps has about $n^2/2$ nodes, that quadruples the work.

Convergence oscillates. Successive values zigzag, because where the strike falls between adjacent terminal nodes changes with the parity of n; in the figure the even-n values sit near 6.79 while the odd-n values drift down toward the limit, and both are still above it at $n = 50$. Reading convergence off a single value of n is therefore unreliable, and at small n adding one step can move the price by more than a one-dollar move in the stock would.

Both are numerical rather than conceptual difficulties, and Chapter 20 addresses them. What matters here is what convergence establishes about the *argument*: the price does not

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

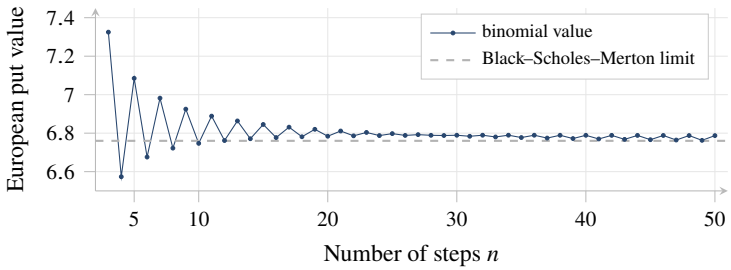


Figure 13.6: European put value against number of steps, $n \geq 3$. $S_0 = 50$, $K = 52$, $\sigma = 30\%$, $r = 5\%$, $T = 2$.

depend on the artificial two-branch assumption, because refining that assumption away leaves the answer essentially unchanged.

Why use a tree at all, then? Not for European values, which have a formula. The tree's worth is that it puts the pricing principle on display (replication, no arbitrage, risk-neutral valuation) and that the principle survives when a decision has to be made at an intermediate date, as in Section 13.7, and a closed-form formula cannot express that.

Why the limit is lognormal

It is worth seeing, without any stochastic calculus, why the tree converges to the model of Chapter 15 rather than to something else. Over n steps the stock price is $S_0 u^j d^{n-j}$, where j is the number of up moves. Taking logarithms,

$$\ln S_T = \ln S_0 + j \ln u + (n - j) \ln d$$

so $\ln S_T$ is a *linear function of* j, and j is a sum of n independent binary draws. The central limit theorem then does the rest: as n grows, j becomes normally distributed, so $\ln S_T$ becomes normal, so S_T becomes **lognormal**.

Two things follow. The lognormal distribution that Chapter 15 assumes is not an extra assumption imposed from outside; it is what any model built from small independent multiplicative shocks produces. And the parameter choice (13.8) was designed to make the variance come out at $\sigma^2 T$ in the limit, so the limiting distribution has exactly the volatility that was put in.

The same reasoning identifies what would have to change to get a different limit. The limit is no longer lognormal if the shocks are not independent, if their size depends on the price level or varies at random, or if they are occasionally very large; the last three are precisely the modifications that Chapters 19 and 21 make to reproduce the volatility smile.

13.10 DIVIDENDS AND OTHER UNDERLYINGS

A known dividend yield. If the stock pays a continuous yield q, the stock itself grows at $r - q$ in the risk-neutral world, so only p changes:

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d} \tag{13.9}$$

Everything else is untouched. The same substitution handles a stock index (with q the dividend yield), a currency (with $q = r_f$, the foreign rate, following Section 5.7), and a futures contract.

Entering a futures contract costs nothing, so in the risk-neutral world a futures price must have zero expected growth. Setting $q = r$ in (13.9) gives

$$p = \frac{1 - d}{u - d} \tag{13.10}$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Table 13.1 collects the cases, and Example 13.8 applies the last of them.

Table 13.1: The probability p for different underlyings.

Underlying	Risk-neutral probability p
Non-dividend-paying stock	$\left(e^{r\Delta t} - d\right)/(u-d)$
Stock or index with yield q	$\left(e^{(r-q)\Delta t} - d\right)/(u-d)$
Currency, foreign rate r_f	$\left(e^{(r-r_f)\Delta t} - d\right)/(u-d)$
Futures contract	$(1-d)/(u-d)$

Example 13.8 An American call on futures, exercised early

A futures price is 100, with $\sigma = 25\%$, $r = 4\%$, and two three-month steps. Value an American call struck at 100.

From (13.8), $u = e^{0.25\sqrt{0.25}} = 1.133148$ and $d = 0.882497$. From (13.10),

$$p = \frac{1 - 0.882497}{1.133148 - 0.882497} = 0.4688$$

Terminal futures prices are 128.40, 100.00, and 77.88, giving payoffs 28.4025, 0, and 0.

Upper node ($F = 113.3148$). Continuation is $e^{-0.01}\,[0.4688 \times 28.4025] = 13.1824$. Exercising yields $113.3148 - 100 = 13.3148$. **Exercise.**

Lower node. Both successors pay nothing; value 0.

Today. Continuation is $e^{-0.01}\,[0.4688 \times 13.3148] = 6.1798$, against an exercise value of 0. Hold.

The American call is worth 6.1798; the European is worth 6.1183. Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early—and here is one exercised early, with no dividend anywhere in the problem.

There is no contradiction, only a limit to that result's scope. The proof relied on the underlying being an asset that must be paid for, so that deferring payment of the strike earns interest. Any positive yield weakens the argument, because waiting now costs the income the underlying pays. A futures contract costs nothing to enter, which is exactly what (13.10) encodes by setting $q = r$: the yield equals the interest rate, and deferring the strike earns nothing net. Early exercise of a deep-in-the-money American call is therefore a live decision on futures options, on currencies with a high foreign interest rate, and on high-yielding indices.

A known cash dividend. A fixed dollar dividend is harder, and the difficulty is worth noting because it is the first place the method's convenience breaks. If the stock drops by D on an ex-dividend date, the tree *stops recombining*: an up-then-down path and a down-then-up path no longer meet, because the dividend is subtracted at different price levels. The node count no longer grows linearly; with a dividend at every step it reaches 2^n. Nothing about the pricing principle changes (the replication argument at each node is untouched), but the computation becomes impractical, and Chapter 20 deals with it.

13.11 PRICING AND HEDGING ARE ONE CALCULATION

One consequence of the argument deserves to be stated on its own, because it is easy to pass over and it governs everything in Part III. The hedge ratio Δ of (13.1) is not a by-product of the pricing calculation. It *is* the pricing calculation: the price was obtained by finding the Δ that makes the portfolio riskless, and the option's value fell out of that construction. There is no separate step in which one first prices the option and then asks how to hedge it.

This is why the sensitivities of an option are not, in this theory, sensitivity analysis. They are the recipe for manufacturing it, and Example 13.5 carried that recipe out node by node. A

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

trader who cannot hedge a position cannot claim to know what it is worth, because the price was defined as the cost of the hedge. Chapter 18 develops the full set of sensitivities on that understanding.

13.12 WHAT THE TREE ASSUMES

The argument of Section 13.1 is valid given its assumptions. They are the assumptions of the continuous-time model the tree converges to, and Section 15.1 lists them and says where each fails: volatility is neither known nor constant, prices jump rather than moving by u or d, and borrowing is not at r. One consequence belongs to the tree. Rebalancing at every node is what makes the hedge riskless, so with transaction costs the imperfection grows with the number of rebalances, and the limit that makes the model converge is also the limit that makes hedging infinitely expensive. Chapter 18 confronts this directly.

None of this makes the model useless. What survives is the *principle* that a derivative replicable from traded assets has a price independent of anyone's beliefs, and that principle is what the rest of Part III builds on.

SUMMARY

A binomial tree lets the underlying move to one of two values over a short interval. Holding $\Delta = (f_u - f_d)/(S_0u - S_0d)$ shares against a written derivative is riskless, so the portfolio must earn the risk-free rate, and that fixes the price. Pricing and delta hedging are one calculation, not two.

The same price is a discounted expectation, $f = e^{-r\Delta t}[p f_u + (1-p) f_d]$ with $p = (e^{r\Delta t} - d)/(u-d)$. The real-world probability of an up move never appears: p is the probability under which the stock earns the risk-free rate, and the device works because the derivative can be replicated, so preferences cancel out. With the discount factor absorbed, the weights are state prices, which are positive exactly when there is no arbitrage; the risk-neutral measure is another name for the absence of arbitrage.

Two states and two traded assets make the market complete, and that is what turns Chapter 11's bounds into a single price. With a third state the option cannot be replicated and the theory delivers only a range; wherever completeness fails, the accurate phrase is "a price, given a model."

Multi-period trees are solved backward. They recombine, so nodes grow linearly, and an American option takes the greater of continuation and exercise value at each node, which prices the early-exercise premium and locates the boundary. With $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$ only volatility must be estimated, and as the steps shrink the terminal distribution becomes lognormal and the value converges, oscillating, to the Black–Scholes–Merton formula. Dividend yields, currencies, and futures change only p; for futures $p = (1-d)/(u-d)$, which is why an American call on futures can be worth exercising early with no dividend anywhere.

The tree assumes costless rebalancing, constant known volatility, no jumps, and borrowing at the risk-free rate, and all four fail in practice. What survives is the principle: a derivative that can be replicated from traded assets has a price independent of anyone's beliefs.

FURTHER READING

The original papers are the place to start: Cox, Ross, and Rubinstein (1979) for the model itself, and Rendleman and Bartter (1979), published independently in the same year. Cox and Ross (1976) is where risk-neutral valuation first appears as a general principle. For convergence behavior, Broadie and Detemple (1996), a comparison of American option valuation methods, documents the odd–even oscillation and sets out which refinements are worth their cost.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

PRACTICE QUESTIONS

13.1 A stock at \$40 will move to \$44 or \$36 in six months. The rate is 5%. Value a European call struck at \$40 by constructing the riskless portfolio, then verify the answer using (13.2).

13.2 In the previous question, compute the risk-neutral probability and verify that the expected stock price under it equals $S_0 e^{rT}$.

13.3 Explain why the real-world probability of an up move does not appear in the option price. Would the answer change if it were 0.9 rather than 0.5?

13.4 A stock at \$100 has $u = 1.15$, $d = 0.87$, and two six-month steps, with $r = 6\%$. Value a European put struck at \$100.

13.5 Repeat the previous question for an American put and compute the early-exercise premium. At which nodes is exercise optimal?

13.6 A stock has volatility 25% a year. Compute u, d, and p for a tree with three-month steps when $r = 3\%$. Verify that $d < e^{r\Delta t} < u$.

13.7 Explain why the binomial value oscillates as n increases, and why the existence of a limit matters for the argument of Section 13.1.

13.8 Show that for a futures contract $p = (1 - d)/(u - d)$, and explain in words why the interest rate disappears.

13.9 A stock at \$50 can move to \$56, \$50, or \$44 over one period, with $r = 4\%$ and $\Delta t = 0.5$. Show that a call struck at \$52 cannot be replicated, and find the range of arbitrage-free prices by imposing positive state prices.

13.10 Explain why a one-period model with two states can be completed by the stock and the bond, and what fails when a third state is added. State what completeness has to do with the option having a single arbitrage-free price rather than a range of them.

13.11 Explain why pricing an option and hedging it are one calculation in a binomial tree rather than two. State where the hedge ratio appears in the pricing arithmetic, and what that implies about a model that prices well but hedges badly.

13.12 State four assumptions the binomial tree makes, and for each name a market condition that violates it. Identify which of the four the convergence result of Section 13.9 is most sensitive to, and why.

FURTHER QUESTIONS

13.13 Example 13.2 computed a 28.5% quarterly required return on the call by working backward from its known price. Repeat this for the put in the same one-step model, and explain why the two required returns differ so sharply in sign and magnitude. What does this tell you about using risk-adjusted discount rates for options?

13.14 A known cash dividend breaks the recombining property. Demonstrate this explicitly with a two-step tree, a \$50 stock, $u = 1.2$, $d = 0.8$, and a \$2 dividend paid between the steps. Then explain why the pricing principle is unaffected even though the computation is not.

13.15 Section 13.12 notes that the limit making the tree converge is also the limit making hedging infinitely expensive under transaction costs. Formalize this: with a proportional cost k per trade, estimate how total hedging cost scales with n, and identify the trade-off between model error and trading cost. (Chapter 18's gamma sets the size of each rebalancing trade.)

13.16 Using a three-step tree, trace the early-exercise boundary for an American put and compare

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

it with the critical price described in Section 11.9. How does the boundary move as you approach expiration, and why must it converge to the strike?

13.17 An analyst prices a convertible bond by valuing the bond and the conversion option separately and adding them. Explain why this is wrong when the bond is callable, and identify which feature of backward induction the additive approach discards.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!